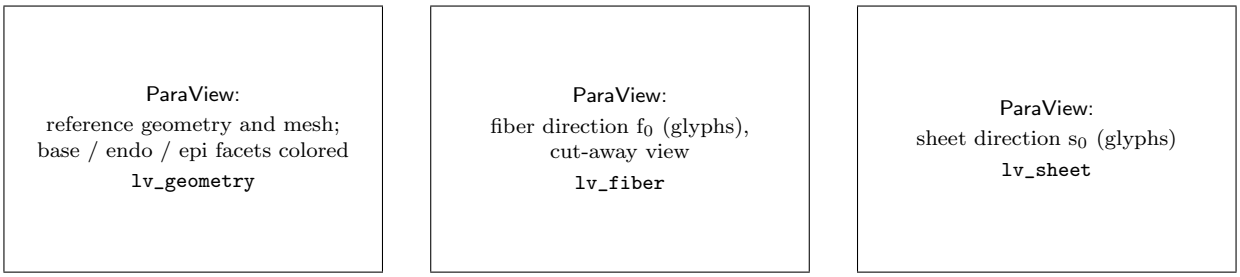
4.6 Dynamic left ventricular problem

A dynamic left-ventricular (LV) problem is finally considered to assess the proposed GLS-stabilized P_1/P_1 formulation in a cardiac-mechanics setting. Compared with the preceding benchmarks, this problem combines a curved three-dimensional geometry, an anisotropic exponential constitutive law with a spatially varying fiber architecture, active fiber contraction, a follower pressure load, and mixed Robin and penalty-type boundary conditions, all under strict incompressibility and strongly time-dependent loading over a full cardiac cycle. The purpose of this example is to demonstrate that the formulation, with the stabilization parameters of Section 3.3 kept unchanged, can follow the large deformation of the ventricle through filling, contraction, and relaxation while maintaining a stable pressure field.

Geometry, mesh, and fiber architecture. The LV is represented by an idealized truncated prolate ellipsoid with the long axis along X_1 (Fig. 20). The endocardial and epicardial surfaces are ellipsoids with semi-axes $(9.0,\, 2.5,\, 2.5)\,\mathrm{cm}$ and $(9.7,\, 3.5,\, 3.5)\,\mathrm{cm}$, respectively, and the base is a conical cut between $X_1 = 2.65\,\mathrm{cm}$ on the endocardium and $X_1 = 2.43\,\mathrm{cm}$ on the epicardium. The reference cavity volume is $V_0 = 167.7\,\mathrm{mL}$. The myocardium is discretized with 21 707 linear tetrahedra and 5840 vertices, i.e., 23 360 velocity–pressure degrees of freedom, with element sizes $h_K \in [2.6,\, 8.2]\,\mathrm{mm}$; the boundary consists of 3590 endocardial, 4638 epicardial, and 430 basal facets. The fiber field follows a rule-based construction: a transmural coordinate $\phi \in [0, 1]$ is obtained from a Laplace problem with $\phi = 0$ on the endocardium and $\phi = 1$ on the epicardium, and the fiber direction f_0 is rotated in the local circumferential–longitudinal plane with a helix angle varying linearly from $+60^\circ$ at the endocardium to -60° at the epicardium. The sheet direction s_0 is aligned with the transmural direction. Both fields are sampled at the element centroids and re-orthonormalized, giving elementwise constant f_0 and s_0 (Fig. 20b,c).



(a) Geometry and mesh (b) Fiber direction f_0 (c) Sheet direction s_0

Figure 20: Dynamic left ventricular problem: idealized geometry, finite element mesh with the basal, endocardial, and epicardial boundaries, and the rule-based fiber and sheet fields.

Constitutive model and active stress. The passive myocardium is described by the incompressible Holzapfel–Ogden model Eq. (19), with the fiber and sheet invariants computed from $\mathbb{C}$ as in Eq. (7) and the positive parts regularized as in Section 3.3. The material parameters are listed in Table 3. A viscous contribution $W_{\mathrm{vis}} = \frac{\eta}{2}\,\mathrm{tr}(\dot{\mathbb{E}}^2)$, with $\dot{\mathbb{E}} = \mathbb{F}^T\,\dot{\mathbb{F}}$ and $\quad = \mathrm{sym}(\dot{\mathbb{F}}\mathbb{F}^{-1})$, and an active fiber stress

$$\mathbb{P}_{\mathrm{act}} = T_a(t)\left[1 + 4.9\left(\sqrt{\bar{I}_{4\mathrm{f}}} - 1\right)\right]\mathbb{F}\,\mathrm{f}_0 \otimes \mathrm{f}_0, \qquad \bar{I}_{4\mathrm{f}} = \mathrm{f}_0 \cdot \bar{\mathbb{C}}\,\mathrm{f}_0, \tag{47}$$

are added to the deviatoric stress, so that the total first Piola–Kirchhoff stress reads $\mathbb{P}_{\mathrm{dev}} + \mathbb{F}\,\eta\,\dot{\mathbb{E}} + \mathbb{P}_{\mathrm{act}} - p\,\mathbb{H}$. These additional stresses enter both the Galerkin terms and the strong momentum residual $\mathbf{R}_{m,h}$ in Eq. (23); the GLS formulation itself is not modified. The effective stiffness in the stabilization parameters is $\mu_{\mathrm{eff}}^{\mathrm{HO}}$ of Eq. (43), including the fiber–sheet coupling term, and the coefficients are those used for all anisotropic examples, $c_t = 1.5$, $c_s = 3.0$, $C_\tau = 0.03$, and $C_c = 0.1$.

Loading and boundary conditions. The cardiac cycle is driven by a prescribed endocardial pressure $p_{\mathrm{endo}}(t)$ and a prescribed active tension $T_a(t)$ (Fig. 21), in kPa and with t in s,

$$p_{\mathrm{endo}}(t) = \begin{cases} 1.067\, t/0.2, & 0 \le t < 0.2, \\ 1.067, & 0.2 \le t < 0.5 \text{ or } t \ge 0.8, \\ 1.067 + 13.46\left[1 - e^{-(t-0.5)^2/0.004}\right], & 0.5 \le t < 0.65, \\ 1.067 + 13.46\left[1 - e^{-(0.8-t)^2/0.004}\right], & 0.65 \le t < 0.8, \end{cases} \tag{48}$$

$$T_a(t) = \begin{cases} 0, & t < 0.5 \text{ or } t \ge 0.8, \\ 84.26\left[1 - e^{-(t-0.5)^2/0.005}\right], & 0.5 \le t < 0.65, \\ 84.26\left[1 - e^{-(0.8-t)^2/0.005}\right], & 0.65 \le t < 0.8. \end{cases} \tag{49}$$

The ventricle is thus passively filled to an end-diastolic pressure of 1.067 kPa (8 mmHg), after which active contraction raises the cavity pressure to 14.5 kPa (109 mmHg) at $t = 0.65$ s before relaxation returns the system to the diastolic state. The boundary conditions are

$$\left(\mathbb{P}\,\mathbf{N}\right)\big|_{\Gamma_{\mathrm{endo}}} = -p_{\mathrm{endo}}(t)\,\mathbb{H}\mathbf{N}, \tag{50}$$

$$\left(\mathbb{P}\,\mathbf{N}\right)\big|_{\Gamma_{\mathrm{epi}}} = -\beta_{\mathrm{epi}}\,(\mathbf{v}\cdot\mathbf{N})\,\mathbf{N}, \tag{51}$$

$$\left(\mathbb{P}\,\mathbf{N}\right)\big|_{\Gamma_{\mathrm{base}}} = -\gamma\left[(\mathbf{u}\cdot\mathbf{e}_1)\,\mathbf{e}_1 + \left(\mathbb{I}_\perp - \hat{\mathbf{R}}\otimes\hat{\mathbf{R}}\right)\mathbf{r}\right], \tag{52}$$

where $\mathbb{P}$ denotes the total stress. Eq. (50) is a follower pressure load acting on the current endocardial surface, and Eq. (51) represents the damping of the surrounding tissue through a Robin-type condition on the normal velocity. In Eq. (52), $\mathbf{e}_1$ is the long-axis direction, $\mathbb{I}_\perp = \mathbb{I} - \mathbf{e}_1\otimes\mathbf{e}_1$, $\mathbf{r}$ is the in-plane position of a basal point relative to the base centre in the current configuration, and $\hat{\mathbf{R}}$ is the corresponding unit radial direction in the reference configuration. The basal penalty therefore suppresses longitudinal motion and circumferential rotation of the base while allowing it to expand and contract radially, which removes the rigid-body modes without clamping the base. All three conditions are natural boundary conditions of the velocity–pressure formulation, and no Dirichlet condition is imposed on the velocity. The system starts from rest in the stress-free reference configuration, $\mathbf{u}_0 = \mathbf{v}_0 = \mathbf{0}$.

The cycle $t \in [0, 1]$ s is integrated with $\Delta t = 5 \times 10^{-3}$ s (200 steps), using backward Euler for the first step and BDF2 thereafter (Section 3.2). Each step is solved by Newton's method with a backtracking line search, absolute and relative residual tolerances 10^{-10} and 10^{-9}, and the MUMPS direct solver. The LV computations were performed with DOLFINx 0.11 and PETSc 3.25; the complete residual, including the stabilization parameters and their dependence on $\mathbb{F}$ through μ_{eff}, is linearized by automatic differentiation.

Table 3: Dynamic left ventricular problem: model and discretization parameters.

Group	Parameter	Value
Holzapfel–Ogden	$a,\ b$	224.49 Pa, 1.6215
	$a_{\mathrm{f}},\ b_{\mathrm{f}}$	2426.7 Pa, 1.8268
	$a_{\mathrm{s}},\ b_{\mathrm{s}}$	556.24 Pa, 0.7746
	$a_{\mathrm{fs}},\ b_{\mathrm{fs}}$	390.52 Pa, 1.695
Viscosity, density	$\eta,\ \rho_0$	100 Pa s, 1000 kg/m^3
Boundary conditions	$\beta_{\mathrm{epi}},\ \gamma$	5×10^3 Pa s/m, 10^6 Pa/m
Fiber architecture	helix angle (endo $\to$ epi)	$+60^\circ \to -60^\circ$
GLS parameters	$c_t,\ c_s,\ C_\tau,\ C_c$	1.5, 3.0, 0.03, 0.1
Discretization	elements / vertices / DOFs	21 707 / 5840 / 23 360
	Δt, steps	5×10^{-3} s, 200

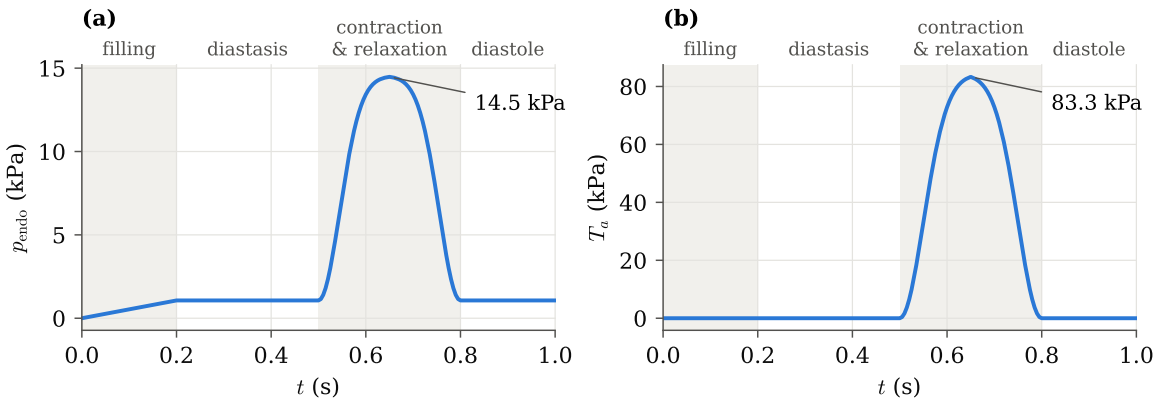


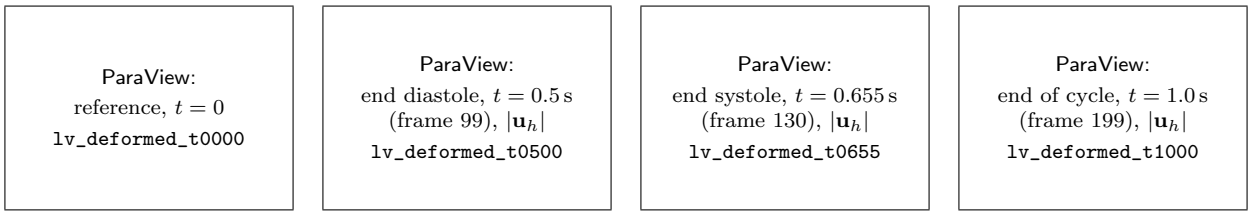
Figure 21: Dynamic left ventricular problem: prescribed (a) endocardial pressure and (b) active tension over one cardiac cycle, Eqs. (48)–(49). Shaded bands mark passive filling and the active phase.

Results. The simulation completes the full cycle without any adjustment of the time step or of the stabilization parameters. Of the 200 steps, 195 converge in two or three Newton iterations (Fig. 25a); at most six iterations are needed at the onset of active contraction ($t = 0.52$–$0.54\,\mathrm{s}$), where T_a and p_{endo} rise most rapidly. The largest final residual norm over the cycle is 2.5×10^{-10}.

Fig. 22 shows the deformed configurations at the reference state, at end diastole ($t = 0.5\,\mathrm{s}$), at end systole ($t = 0.655\,\mathrm{s}$, minimum cavity volume), and at the end of the cycle. The deformation is large throughout. During passive filling the cavity volume increases from 167.7 mL to 394.9 mL, the long-axis length increases by 27 % (from 122.3 mm to 155.1 mm), and the maximum displacement reaches 34.5 mm, about 28 % of the reference long-axis length. Active contraction then reduces the cavity volume to 145.7 mL, corresponding to an ejection fraction of 63.1 % (Fig. 23a). The contraction exhibits the characteristic kinematics of the ventricle: relative to end diastole, the long axis shortens by 11.6 % and the equatorial wall thickens by 62 % (from 6.3 mm to 10.2 mm, Fig. 23c). The pressure–volume loop in Fig. 23b displays the filling, contraction, ejection, and relaxation phases of the cycle.

Fig. 24 presents the pressure field p_h on the deformed configurations. The pressure remains smooth across the wall at all stages of the cycle, without the element-scale checkerboard patterns of the unstabilized equal-order discretization observed in the preceding benchmarks. Its extreme values stay bounded and follow the applied cavity pressure (Fig. 25c), with $p_h \in [-2.07, 8.17]\,\mathrm{kPa}$ over the cycle compared with the peak endocardial pressure of 14.5 kPa. Since the incompressibility constraint is imposed in the rate form of Eq. (20c), the volume of the myocardial wall is preserved in an integral sense: the mean Jacobian $\bar{J} = |\Omega_0|^{-1} \int_{\Omega_0} J_h \, \mathrm{d}\Omega$ deviates from unity by at most 0.20 % over the cycle (Fig. 25b), and the deviation arises almost entirely during the active phase.

These results show that the proposed GLS-stabilized P_1/P_1 formulation, with the effective stiffness $\mu_{\mathrm{eff}}^{\mathrm{HO}}$ as the only model-dependent ingredient of the stabilization parameters, carries over directly to an anisotropic, actively contracting left ventricle with follower, Robin, and penalty-type boundary conditions. It robustly captures the large global deformation through a complete cardiac cycle while maintaining a stable pressure field.



(a) $t = 0$ (b) $t = 0.5\,\mathrm{s}$ (ED) (c) $t = 0.655\,\mathrm{s}$ (ES) (d) $t = 1.0\,\mathrm{s}$

Figure 22: Dynamic left ventricular problem: deformed configurations colored by the displacement magnitude $|\mathbf{u}_h|$ at the reference state, end diastole, end systole, and the end of the cycle. The gray outline indicates the reference configuration.

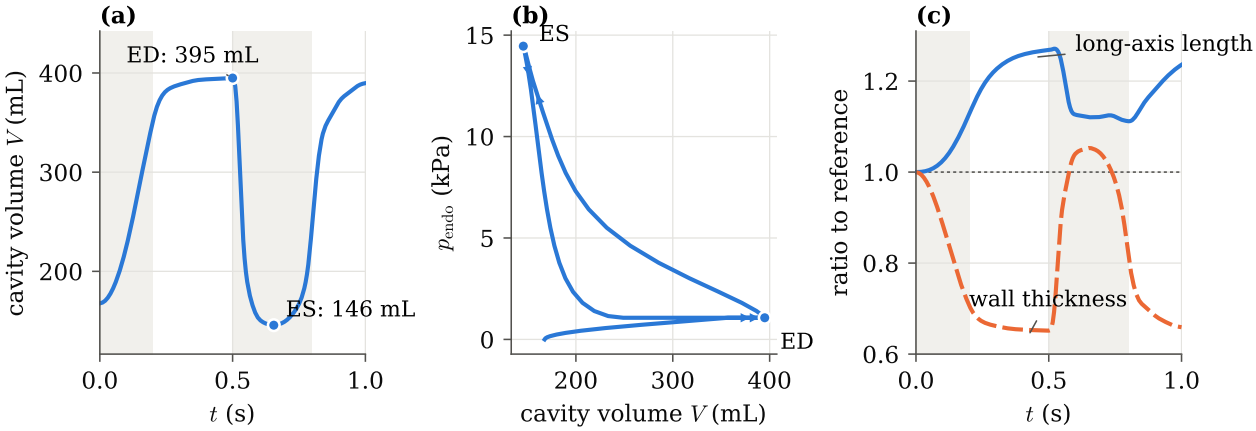
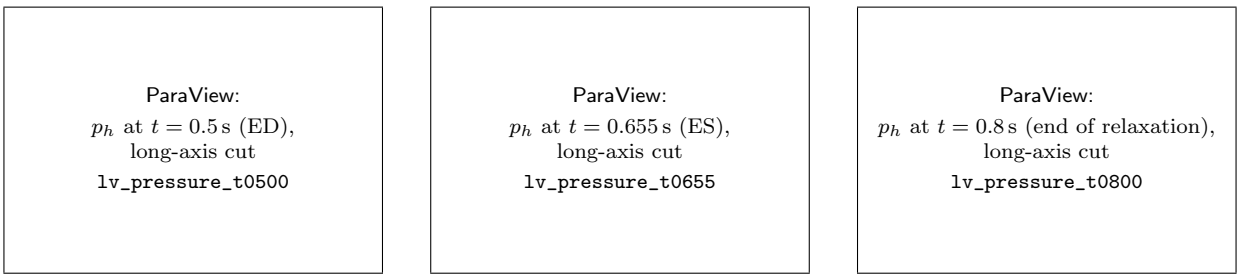


Figure 23: Dynamic left ventricular problem: global response. (a) Cavity volume, with end diastole (ED) and end systole (ES) marked; (b) pressure–volume loop; (c) long-axis length and equatorial wall thickness relative to the reference configuration. The cavity volume is computed exactly from the discrete endocardial surface as $V = -\frac{1}{3}\int_{\Gamma_{\mathrm{endo}}} (\mathbf{x} - \mathbf{x}_0)\cdot\mathbf{n}\,da$, with $\mathbf{x}_0$ on the basal plane of the endocardium.



(a) $t = 0.5\,\mathrm{s}$ (ED) (b) $t = 0.655\,\mathrm{s}$ (ES) (c) $t = 0.8\,\mathrm{s}$

Figure 24: Dynamic left ventricular problem: pressure field p_h on the deformed configuration at end diastole, end systole, and the end of relaxation, shown on a long-axis cut through the wall.

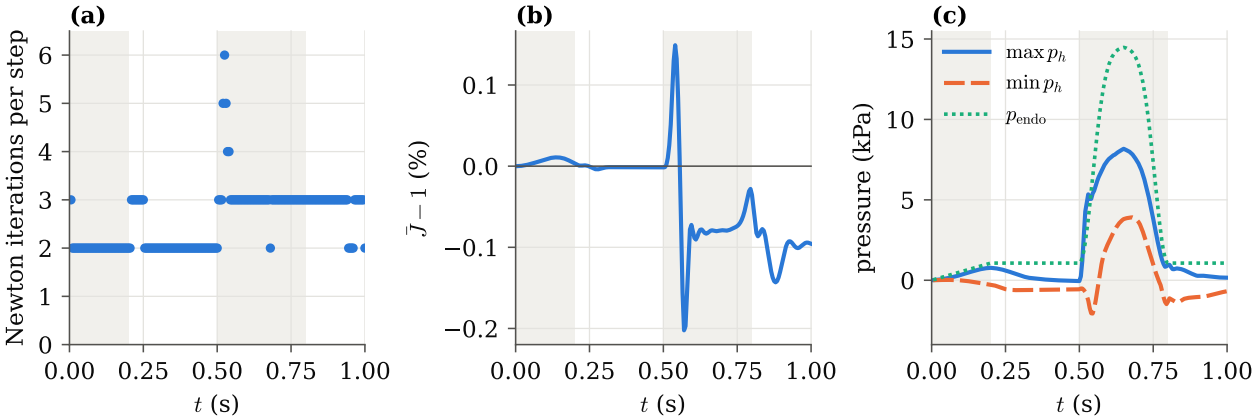


Figure 25: Dynamic left ventricular problem: (a) Newton iterations per time step; (b) deviation of the mean Jacobian $\bar{J}$ from unity; (c) extreme values of the pressure field compared with the prescribed endocardial pressure.